

GlycanBench: Tool-by-Tool State-of-the-Art Comparison

Each row corresponds to one entry in the application menu (as defined in `NavBar.tsx` and routed in `App.tsx`). The **State of the Art** column describes what the field currently offers for that task. The **Our Contribution** column describes what GlycanBench implements, verified against the actual source code, including precise algorithmic and UX details.

CREATE

Tool	State of the Art	Our Contribution
Glycan Molecule (<code>/GlycanMolecule</code>)	Standard cheminformatics pipelines generate 3D conformers from SMILES using RDKit's distance-geometry algorithm (ETKDG) followed by MMFF94 or UFF force-field minimization. Tools such as OpenBabel and Avogadro expose this capability locally. Carbohydrate-specific force fields (GLYCAM06, GROMOS 56A6CARBO) are the gold standard for conformational accuracy but require expert setup, local installation, and are not web-accessible for arbitrary inputs. GLYCAM-Web covers only a fixed library of	GlycanBench provides a visual click-to-build glycan constructor in the browser: users assemble sequences by clicking sugar tokens (60+ monosaccharides), linkage tokens, and branching brackets [] from validated button palettes. Structural grammar is enforced in real time (a bond must follow a sugar; a sugar must follow a bond or bracket). On submission, the built sequence is sent to three simultaneous backend calls: (1) <code>POST /api/draw</code> — SNFG 2D rendering via <code>GlycoDraw</code> ; (2) <code>POST /api/convert</code> — format conversion to IUPAC, SMILES, GlycoCT, WURCS; (3) <code>POST /api/visualize</code> — IUPAC canonicalization via <code>glycowork</code> , SMILES via <code>IUPAC_to_SMILES()</code> , ETKDGv3 3D embedding (chirality enforcement, small-ring torsion corrections), minimized with MMFF94s (static variant, better for cyclic systems) with UFF fallback, MDL Molfile returned and rendered in 3Dmol.js with four switchable display styles (Ball & Stick, Spacefill/CPK, Wireframe, Stick Figure) and screenshot export. The integration of a guided sequence builder, simultaneous 2D/3D/format output, and a tiered ETKDGv3 + MMFF94s + UFF optimization pipeline in a browser-native 3D viewer is not available in any existing glycomics web tool.

Tool	State of the Art	Our Contribution
	pre-computed structures.	
Biosynthetic Networks (/BiosyntheticNetworks)	glycowork (Thomès et al., 2021–2024) introduced programmatic biosynthetic network construction from sets of IUPAC-condensed glycans using curated enzymatic rules. GlyConnect and GlyTouCan maintain curated biosynthetic data as static databases. No web application exposes interactive, fully parameterized biosynthetic network construction from arbitrary user-supplied glycan sets.	GlycanBench wraps <code>glycowork.network.biosynthesis.construct_network()</code> in a REST API and an interactive Cytoscape.js frontend. Users control three orthogonal parameters: (1) edge type (<code>monolink</code> , <code>full_reaction</code> , or enzyme -labeled); (2) allowed PTMs — dynamically fetched from <code>lib.ptm_dict</code> at startup and toggled individually (acetylation, phosphorylation, sulfation, etc.); (3) permitted roots — comma-separated custom reducing-end anchors (defaulting to <code>Gal(b1-4)GlcNAc-o1</code> and <code>Gal(b1-4)Glc-o1</code>). The response is a Cytoscape.js element list with <code>is_input</code> and <code>is_root</code> node flags. The frontend supports dark/light mode, debounced node search with zoom-to-result animation, node tap highlighting (selected node + neighbors + connected edges), PNG export at 2× resolution, <code>breadthfirst</code> directed layout with animated re-layout, and structured metadata (node/edge count, params used). Four curated example glycan sets (complex N-glycans, fucosylated oligosaccharides, GSL cores, sulfated/phosphorylated glycans) are provided. The combination of all three parameter axes with a live Cytoscape.js interactive graph in a web interface is not available elsewhere.
Interconvert Between Formats (/GlycanFormatConverter)	GlycoWorkbench and the <code>glypy</code> library support bidirectional conversion among GlycoCT, WURCS, and IUPAC. <code>glycowork</code> provides IUPAC canonicalization and IUPAC→SMILES. No single web endpoint handles all four major formats (IUPAC, GlycoCT, WURCS, SMILES) in one request while	GlycanBench accepts IUPAC-condensed, GlycoCT, or WURCS as input and returns all four representations in a single API response. The IUPAC path uses <code>glycowork.motif.processing.canonicalize_iupac()</code> + <code>IUPAC_to_SMILES()</code> . GlycoCT/WURCS paths use <code>glypy.io</code> loaders and dumpers. A known limitation — IUPAC→GlycoCT and IUPAC→WURCS — is exposed to the user with an explicit "not supported" message rather than silently failing or omitting the field. The frontend filters unsupported/failed fields from the results panel and shows them separately as orange-labeled notices. Per-format copy-to-clipboard and file download (with correct extensions: <code>.gct</code> , <code>.wurcs</code> , <code>.txt</code>) are provided. This transparent handling of partial conversion coverage, alongside syntax-highlighted collapsible result cards, is a practical UX contribution over existing command-line tools.

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	clearly reporting unsupported conversion paths.	

VISUALIZE

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2D Draw (/GlycanDrawer)	The SNFG (Symbol Nomenclature for Glycans) standard (Varki et al., 2015; NIBIB 2017) is the community-adopted 2D depiction scheme. DrawGlycan-SNFG and GlycoWorkbench are reference tools. glycowork's GlycoDraw function (Thomès et al., 2021) provides programmatic SNFG rendering with optional motif highlighting, built on Matplotlib. Most standalone tools require local installation or a Java runtime.	GlycanBench exposes GlycoDraw(draw_this, highlight_motif) as a REST endpoint (POST /api/draw). Any substring of the IUPAC string can be passed as highlight_motif, triggering selective color overlay on matching structural elements — a feature absent in most web-based SNFG renderers. The result is a base64-encoded PNG returned in one call, directly renderable in the browser with a download link. The "Autofill Example" button prefills a complex sialylated, fucosylated N-glycan with a Lewis-motif highlight to demonstrate the feature. This is the only web-accessible SNFG renderer providing per-motif highlighting in a single stateless request.
3D Representation (/visualize)	Browser-based 3D molecular viewers (3Dmol.js, NGL Viewer, Speck) are mature. GLYCAM-Web provides GLYCAM06-minimized 3D structures but only for its pre-computed library. No web service generates high-quality, on-demand 3D structures from IUPAC strings for arbitrary glycan inputs with force-field minimization optimized for ring-heavy carbohydrate systems.	GlycanBench generates a 3D conformer for any syntactically valid IUPAC input on demand (POST /api/visualize) using a multi-step pipeline: ETKDGv3 (RDKit's best distance-geometry algorithm, with chirality enforcement and small-ring torsion corrections active — directly beneficial for pyranose rings) with classic ETKDG as fallback; minimization with MMFF94s (the static variant, better suited to cyclic systems than standard MMFF94) for up to 2000 iterations with a second pass if energy remains high; UFF fallback when MMFF94s cannot be set up for the molecule. The result is an MDL Molfile rendered in 3Dmol.js with a dark background, semi-transparent VDW surface overlay, four display style buttons (Ball & Stick, Spacefill, Wireframe, Stick Figure), Reset View, Save as PNG, and Clear & Reset. The absence of a pre-computed library restriction — any valid IUPAC string triggers on-demand generation — combined with the ETKDGv3 + MMFF94s + UFF tiered optimization pipeline, is the key differentiator from GLYCAM-Web and generic cheminformatics tools.
KEGG Pathway View	KEGG (Kanehisa et al.) is the gold standard for	GlycanBench implements three backend endpoints: (1) GET /api/pathway?kegg_id=... — tries

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(/pathwayMaps)	metabolic and glycan biosynthesis pathway maps. The KEGG REST API provides programmatic image access. Browser requests to <code>rest.kegg.jp</code> fail due to CORS restrictions, preventing direct client-side rendering. Most bioinformatics tools (Bioconductor's KEGGREST , PathVisio) consume KEGG data locally.	<code>rest.kegg.jp/get/{id}/image</code> first, falls back to the direct <code>kegg.jp/kegg/pathway/...</code> PNG URL pattern, returns the validated image URL; (2) <code>GET /api/search_pathways?query=...</code> — a fully compliant FastAPI endpoint using a <code>Query</code> parameter (<code>min_length=3</code>) that proxies the KEGG <code>find/pathway/</code> endpoint for autocomplete search (debounced at 300 ms client-side), returning structured <code>{id, name}</code> results with proper <code>HTTPException</code> error handling; (3) <code>GET /api/proxy_image?url=...</code> — downloads the KEGG PNG server-side and streams it as a file download, bypassing browser CORS restrictions. The frontend uses <code>react-zoom-pan-pinch</code> for interactive pan/zoom/reset of the pathway image. Ten curated example pathways (N-glycan biosynthesis, O-glycan biosynthesis, heparan sulfate, ganglio-series GSLs, glycan degradation, LPS biosynthesis, etc.) are provided as quick-access buttons. This is the only glycomics workbench embedding live KEGG pathway browsing with integrated autocomplete search and CORS-bypassed download.

ANALYSE

Tool	State of the Art	Our Contribution
Monosaccharide (characterization, /characterize)	Cross-species statistical analysis of monosaccharide occurrence is available programmatically via <code>glycowork.motif.analysis.characterize_monosaccharide()</code> . Manual cross-queries across GlyTouCan, UniCarbKB, and GlycoStore are the alternative. No interactive web tool provides taxonomic-rank-stratified occurrence charts for arbitrary monosaccharides on demand with user-controlled parameters.	GlycanBench v1.0.0 provides a <code>characterize_monosaccharide()</code> endpoint with parameters: <code>rank</code> , <code>focus</code> , <code>thresh</code> in a <code>dict</code> . Users control: <code>tax</code> (free text), <code>tax</code> through <code>Variant</code> dropdown), <code>bio</code> (e.g. "Bacteria") <code>chemical</code> modifications (1–50, <code>min_inclusion</code>). The quality Matplotlib base64 PNG. <code>tax</code> <code>Rha</code> , <code>rank=Kin</code> <code>modifications=</code> meaningful glycan parameterized, with real-time feedback on underlying statistics. <code>glycowork</code> .
Glycan (insight, /GlycanInsight)	Annotation of a glycan's biological context requires manual cross-queries of GlyTouCan, UniCarbKB, GlyConnect, and the literature. <code>glycowork</code> 's <code>get_insight()</code> automates this against	GlycanBench v1.0.0 provides a <code>get_insight()</code> endpoint that returns a condensed sequence of GlyTouCan accession numbers.

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	its internally curated database. No web tool presents all annotation axes (species, phyla, motifs, GlyTouCan ID, cell lines, diseases) in a single structured dashboard.	detected: starts GlyTouCan IDs via glytoucan.get_class() + get_insight(). regex-parsed in The frontend re animated dash chart (Chart.js) species tag clo toggle at 40 en table with up/d coding; a cell li and a GlyTouC biological infor empty state dis no database er The structured from a single te any current gly
Molecule Descriptors (/DescriptorCalculator)	Cheminformatics descriptor calculation is mature (RDKit, Mordred, PaDEL). For glycans specifically, no dedicated web server computes the full physicochemical descriptor set together with glycan-specific SMARTS motif counts and five fingerprint types in a unified request. glycowork computes glycan-specific graph descriptors but not cheminformatics fingerprints.	GlycanBench c (canonicalized SMILES: 17 pr descriptors (N atom count, for stereocenters, stereocenters, aliphatic/satur donors/accept TPSA, LogP, fr refractivity); ful with O/N ratio; SMARTS moti [0X2r5; !R0] , [0X2r4; !R0] , [CX3](=O) , c groups); and 5 (Morgan R2, M Topological Tor maxPath=7). A SMILES enable across all finge request. The fr 8 color-coded, (Molecular Wei & H-Bonding, / Physicochemic Glycan Motifs, with animated i a PubChem loc download. This specific SMAR standard chem

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		panel in one page not available elsewhere
Motif Mutation (/MotifMutation)	Systematic in-silico mutagenesis of glycan sequences to study motif robustness has not been exposed as a web tool in any published glycomics platform. Glycan mutagenesis has been performed in ad hoc research code.	GlycanBench tool by splitting on [monosaccharide] performs stochastic mutagenesis with intensity mode: <code>n_mut position (n_mut+1 , cap length), extreme positions</code> . For sequences (consider all pentamer glycan windows: sugar, step 2) : frequencies table ensemble. Rep monosaccharide monosaccharide (experimentally bonds from a high 35 α/β linkages the frequency color bar chart, color (green/yellow/red) with 5 collapse explains the pie interactive tool silico glycan motif frequency profile robustness.

COMPARE

Tool	State of the Art	Our Contribution
Two Glycans (/CompareGlycans)	Glycan similarity metrics include tree edit distance (used in GlycoSimilarity), substructure-based overlap (GlycoMod), glycowork graph similarity, and fingerprint-based Tanimoto similarity. No interactive web tool reports multi-fingerprint Tanimoto comparisons simultaneously for a	GlycanBench first converts both IUPAC inputs to SMILES via <code>POST /api/convert</code> , then computes <code>POST /api/compare_glycans</code> — pairwise Tanimoto similarity across any user-selected subset of five 2048-bit fingerprint types: Morgan R2 (ECFP4-equivalent), Morgan R3 (ECFP6-equivalent), Atom Pair, Topological Torsion, RDKit path (maxPath=7). A "Select All" toggle is provided. Results are displayed with the derived SMILES strings shown explicitly, and each fingerprint score is accompanied by an <code>IoHelpCircleSharp</code> hover tooltip explaining what structural features that fingerprint encodes. Presenting five fingerprint scores

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	glycan pair with per-fingerprint hover explanations.	simultaneously enables users to assess whether similarity is consistent across fingerprint families (robust structural resemblance) or fingerprint-dependent (only specific substructures shared). This multi-fingerprint consensus view with per-fingerprint explanations is not available in any existing glycan comparison tool.

ALIGN

Tool	State of the Art	Our Contribution
Glycan Sequences (/sequenceAlignment)	Pairwise alignment of glycan sequences has been approached via BLAST-like string alignment (inappropriate for branched structures), SweetAlign (2006, Smith-Waterman with simple match/mismatch scoring), and custom solutions. GLYSUM (Alocci et al., 2015, <i>Glycobiology</i>) introduced the first rigorously derived glycan substitution scoring matrix, analogous to BLOSUM/PAM for amino acids, encoding evolutionary and structural relationships between monosaccharides and linkages. BioPython's <code>PairwiseAligner</code> supports custom substitution matrices but has never been exposed in any glycan-specific web application.	GlycanBench implements global Needleman-Wunsch alignment (BioPython <code>PairwiseAligner</code> , <code>mode=global</code>) using GLYSUM.xlsx as the default substitution matrix — the only glycan-native scoring scheme in the published literature. IUPAC strings are tokenized into alternating [monosaccharide, linkage] arrays; <code>difflib.get_close_matches</code> (cutoff 0.85) maps near-exact names to the vocabulary rather than hard-failing; unknown sugars → UNK1, unknown linkages → UNK2 (both with zero substitution score). Affine gap penalties (configurable open + extend) apply in both modes. Users may switch between GLYSUM and custom match/mismatch/gap scoring via radio buttons, with greyed-out score fields when GLYSUM is active. Output: token-padded columnar alignment display (all tokens padded to uniform width, pipe `

CLUSTER

All three cluster tools use the same backend endpoint (`POST /api/cluster/run`) with the `mode` parameter (`standard`, `optimal_k`, `outliers`) and `clustering_method` (`agglomerative`, `kmeans`) controlling behavior. All three share an identical UI pattern: a glycan name+IUPAC input table with add/remove rows, magic-fill defaults, and an advanced options collapsible panel (fingerprint type, radius, `n_bits`, linkage method, distance metric). The `iupac` string is now preserved through the normalization pipeline alongside the derived SMILES, enabling glycan-native distance metrics that require the original symbolic sequence.

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Cluster Three or More Glycans (/cluster/multiple)	Glycan clustering has been performed using tree edit distance ($O(n^2)$, computationally expensive), glycowork graph similarity, and SMILES-fingerprint Tanimoto distance with agglomerative clustering. SciPy linkage / fcluster is the standard implementation. Seaborn heatmaps are standard for pairwise distance visualization. No web tool provides interactive agglomerative glycan clustering with both fingerprint-based and glycan-native graph-similarity metrics selectable at runtime.	IUPAC strings are batch-converted to SMILES via <code>glycowork.motif.processing.IUPAC_to_SMILES()</code> or accepted directly, with the original IUPAC string retained for glycan-native distance computation. Users select: clustering method (Agglomerative with distance threshold slider, or K-means with k slider); fingerprint type (Morgan, Atom Pair, Topological Torsion, RDKit); radius (2 or 3); n_bits (1024 or 2048); linkage method (average/UPGMA, single, complete, ward — ward auto-forces Euclidean metric); distance metric — Tanimoto, Dice, Cosine, Euclidean (all fingerprint-based), or glycowork graph similarity (<code>glycowork.motif.graph.compare_glycans()</code> , bypasses fingerprints entirely, uses the original IUPAC strings directly). The <code>glycowork_similarity_available</code> flag in the response signals to the frontend whether this metric can be offered. Outputs: Matplotlib dendrogram (agglomerative only), Seaborn pairwise heatmap, cluster membership dict, PNG download, and CSV of cluster assignments. K-means uses <code>sklearn.cluster.KMeans</code> (<code>n_init=10</code> , <code>max_iter=300</code> , <code>random_state=42</code>) on the raw fingerprint bit matrix. The addition of a glycan-native graph-similarity metric alongside the full suite of cheminformatics fingerprints, switchable at runtime without code changes, is unique to GlycanBench.
Optimize Number of Clusters (/cluster/optimize)	Determining the optimal number of clusters for glycan data has no established dedicated tool. Elbow plots (inertia vs. k) and silhouette analysis are standard in general cheminformatics. For hierarchical clustering, sweep plots of cluster count vs. distance threshold are used ad hoc in research code but not exposed interactively for glycan data.	GlycanBench runs <code>mode=optimal_k</code> : agglomerative clustering is executed at each threshold in a user-configurable list (default: 0.1, 0.2, ..., 0.7 in steps of 0.1), recording cluster count at each threshold. The glycowork graph metric is available here as well. Outputs: a Matplotlib line plot of cluster count vs. threshold (the "elbow plot"), a numeric threshold-curve table rendered in the UI, the cluster dendrogram for the chosen starting threshold, and the pairwise distance heatmap. All images are PNG-downloadable. This interactive, glycan-specific threshold-sweep visualization does not exist in any current glycomics tool.
Detect Outlier Glycans (/cluster/outliers)	Outlier detection in glycan sets has been performed manually by dendrogram inspection. No automated, web-	GlycanBench runs <code>mode=outliers</code> : after agglomerative or k-means clustering, any glycan forming a singleton cluster (size = 1) is flagged as a potential outlier. For each singleton, mean pairwise distance to all other glycans is computed from the full distance matrix as a severity score (works with any metric including the glycowork graph

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	accessible tool reports singleton clusters with quantitative outlier severity for glycan data.	similarity). The UI renders outliers in a red-highlighted list with cluster ID and mean distance. Singletons with mean distance > 0.45 receive an additional "STRONG OUTLIER" badge. A green "no outliers detected" message is shown when all clusters have size ≥ 2 . The dendrogram and heatmap are also returned to visually confirm outlier separation. This is the first automated, quantitative glycan outlier detection tool accessible without programming.

PREDICT

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Immunogenicity (/prediction)	Glycan immunogenicity prediction is an emerging area. GlyNet and related GNN approaches (Bojar et al., 2021, <i>Cell Chemical Biology</i>) use graph representations with GIN or GAT architectures. glycowork provides LSTM and GNN classifiers for glycan property prediction. The field consensus is that graph-based representations capturing branched topology outperform linear sequence models. Known xenoantigens (AlphaGal = Gal(α 1-3)Gal, Neu5Gc) are well-established. No web-accessible immunogenicity predictor for arbitrary glycan sequences currently exists in the public domain.	GlycanBench deploys a glycword-graph MPNN classifier. The representation pipeline: IUPAC $\rightarrow$ regex normalization (unusual bond notation) $\rightarrow$ tokenization into alternating [sugar, linkage] tokens $\rightarrow$ unknown-token masking (UNK_SUG , UNK_BOND) $\rightarrow$ glycword extraction (5-token pentamer windows, step 2: sugar–bond–sugar–bond–sugar, analogous to k-mers) $\rightarrow$ vocabulary lookup in glycword_vocab.json $\rightarrow$ graph construction (nodes = glycword indices, forward+reverse sequential edges) $\rightarrow$ torch_geometric Data . MPNNClassifier architecture: `nn.Embedding(

CHAT

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Chat (/GlycomicsChat)	General-purpose LLMs (GPT-4, Claude, Gemini) have broad scientific knowledge but no glycomics specialization, no live database access, and no domain enforcement. Domain-specific	GlycanBench deploys a Groq-hosted openai/gpt-oss-120b (temperature=0.1, top_p=0.9, max_tokens=2048) behind a ~200-line glycomics system prompt defining the model as a "Glycan Discovery Engine." Domain gating is strict: non-glycomics queries receive a canned refusal. External tools (user-selectable via category toggles in the UI): Literature — PubMed via LangChain PubmedQueryRun ; ArXiv via ArxivAPIWrapper (top_k=5); Databases — GlyTouCan REST API (auto-triggered when regex <code>G\d{5}[A-Z]{2}</code> matches the input, fetching WURCS, IUPAC, mass, formula); structure analysis and synthesis pathway reasoning handled in-prompt.

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	chatbots using RAG over glycomics literature exist in research settings but are not publicly accessible and do not integrate live tool calls. LangChain ReAct agents are state of the art for LLM + external tool integration. No publicly available glycomics chatbot combines a domain-enforced LLM with simultaneous live PubMed, ArXiv, and GlyTouCan database access and a structured expert persona.	Intelligent tool selection: when a category is selected, <code>intelligent_tool_selection()</code> now uses an LLM-based router — a compact JSON-returning routing prompt sent to the same Groq model asks which specific tools are genuinely useful for the query. The parsed result overrides individual tool flags, with guaranteed fallbacks (PubMed for Literature, Structure Analysis for Databases) if no tool is selected by the router. A keyword-scoring heuristic (<code>_keyword_route_tools</code>) is preserved as an automatic fallback when the LLM routing call fails. Question-type analysis (<code>analyze_question_type</code>) categorizes the query (mechanistic / structural / functional / current research / methodological / clinical / biosynthetic) and injects the category as context into the prompt. Confidence scoring (<code>calculate_response_confidence</code>) evaluates response length, scientific term density, glycan-specific notation presence, quantitative content, external evidence used, and uncertainty acknowledgment. The frontend renders tool-used badges, confidence badges, processing time badges, and — when a GlyTouCan accession is resolved — structured panels for WURCS, IUPAC, GlycoCT, mass, and molecular formula. The combination of strict glycomics domain enforcement, LLM-driven tool routing with keyword fallback, live multi-source retrieval, question-type-aware prompt augmentation, and a fully typed React chat interface is not available in any existing tool.